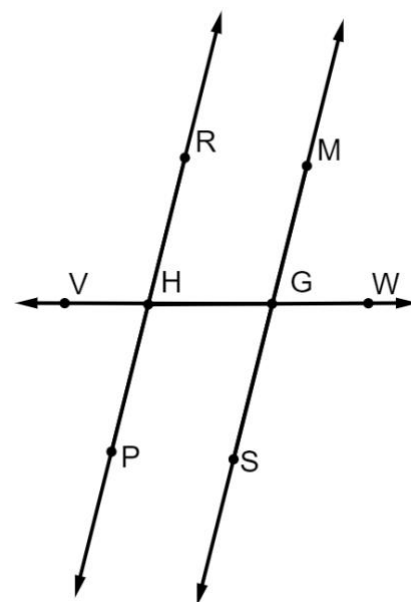


1. Lines RP , VW , and MS are shown. $\overleftrightarrow{VW}$ intersects $\overleftrightarrow{RP}$ at point H . $\overleftrightarrow{VW}$ intersects $\overleftrightarrow{MS}$ at point G .

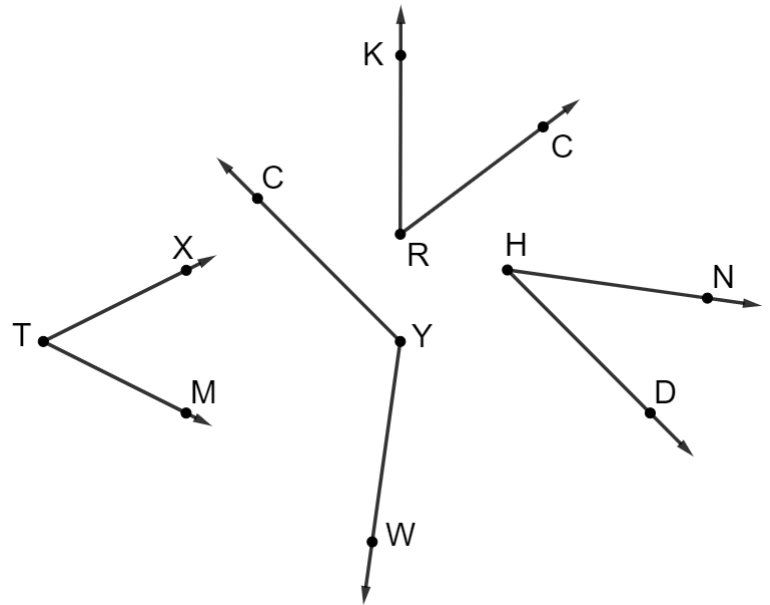


First, determine if each condition is sufficient on its own to show that $\overleftrightarrow{RP} \parallel \overleftrightarrow{MS}$. Then, write a justification.

	Condition	Sufficient to show that $\overleftrightarrow{RP} \parallel \overleftrightarrow{MS}$?	Justification
a.	$\angle PHG \cong \angle MGH$	☐ Yes ☐ No	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
b.	$\angle RHG$ and $\angle MGH$ are supplementary.	☐ Yes ☐ No	If two lines are intersected by a transversal and the consecutive interior angles are supplementary, then the lines are parallel.
c.	$\angle SGW \cong \angle PHV$	☐ Yes ☐ No	The consecutive exterior angles must be supplementary angles to apply the converse of the consecutive exterior angles' theorem.
d.	$m\angle WGM \cong m\angle GHR$	☐ Yes ☐ No	If two lines are intersected by a transversal and the corresponding angles are congruent, then the lines are parallel. If two angles are congruent by definition of congruence, they have the same measure.

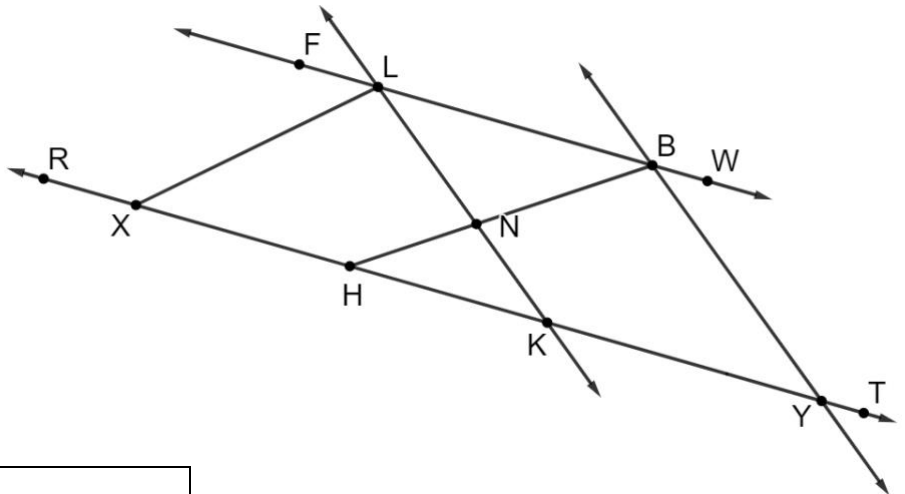
2. Four angles are shown where $\angle NHD$ is complementary to $\angle XTM$, $\angle KRC$ is supplementary to $\angle CYW$, and $\angle KRC$ is complementary to $\angle NHD$. $m\angle CYW = 11x - 27$, $m\angle XTM = 3x + 11$, and $m\angle NHD = 2x + 9$.

Select all of the statements that are true based on the given information.



- ☒ $m\angle XTM = m\angle KRC$
- ☒ $3x + 11 + 2x + 9 = 90^\circ$
- ☐ $11x - 27 + 2x + 9 = 180^\circ$
- ☒ $m\angle XTM + m\angle CYW = 180^\circ$
- ☐ $11x - 27 + 3x + 11 + 2x + 9 = 180^\circ$

3. A diagram is shown where $\overleftrightarrow{LK} \parallel \overleftrightarrow{BY}$, $\overleftrightarrow{FW} \parallel \overleftrightarrow{XT}$, $m\angle XLN = 100^\circ$, $m\angle NHK = 34^\circ$, and $m\angle YBH = 107^\circ$.



Complete the table.

a.

$$m\angle KYB = 39^\circ$$

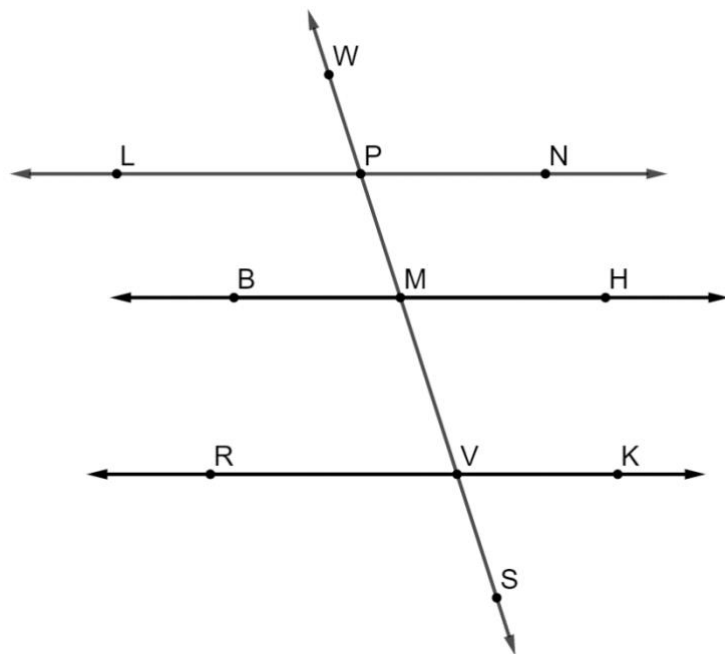
b.

$$m\angle XLF = 41^\circ$$

4. Parallel lines $\overleftrightarrow{LN}$, $\overleftrightarrow{BH}$, and $\overleftrightarrow{RK}$, with a transversal $\overleftrightarrow{PW}$, are shown $m\angle BMV = 108^\circ$ and $m\angle KVS = 72^\circ$.

Complete the table.

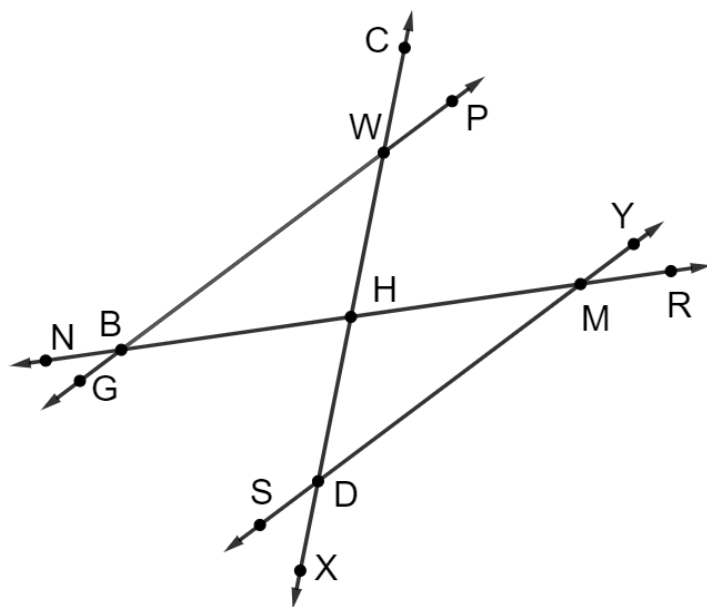
a.	$m\angle WPN = 108^\circ$
b.	$m\angle LPW = 72^\circ$



5. Parallel lines $\overleftrightarrow{BW}$ and $\overleftrightarrow{DM}$ are shown where $m\angle XDM = 138^\circ$ and $m\angle HBW = 29^\circ$.

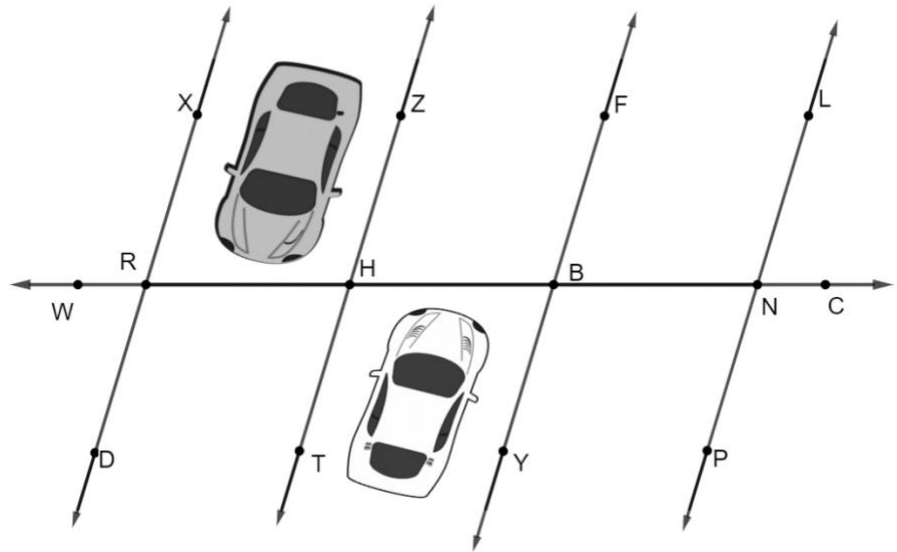
Complete the table.

a.	$m\angle CWB = 138^\circ$
b.	$m\angle HMY = 151^\circ$



6. An engineer is designing a parking lot for a local grocery store. The parking spaces are marked with lines where $\overleftrightarrow{XD} \parallel \overleftrightarrow{TZ} \parallel \overleftrightarrow{YF} \parallel \overleftrightarrow{LP}$ where $\overleftrightarrow{WC}$ is a transversal, as shown, where $m\angle CNL = (6x - 17.6)^\circ$, $m\angle FBH = (12y - 22)^\circ$, and $m\angle THR = (3x + 29.2)^\circ$.

- a. Determine the value of x .



$x =$	15.6
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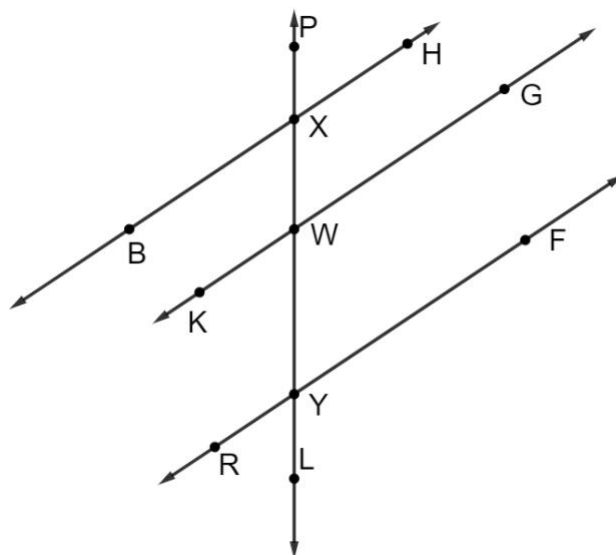
- b. Determine the value of y .

$y =$	10.5
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7. A diagram is shown where $\overleftrightarrow{PL}$ intersects lines BH , KG , and RF .

Given: $\overleftrightarrow{BH} \parallel \overleftrightarrow{RF}$, $\angle HXW \cong \angle YWG$

Prove: $\overleftrightarrow{RF} \parallel \overleftrightarrow{KG}$



Complete the proof.

Statement	Reason
1. $\overleftrightarrow{BH} \parallel \overleftrightarrow{RF}$	1. Given
2. $\angle HXW \cong \angle LYF$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle HXW \cong \angle YWG$	3. Given
4. $\angle LYF \cong \angle YWG$	4. Transitive property
5. $\overleftrightarrow{RF} \parallel \overleftrightarrow{KG}$	5. If two lines and a transversal form corresponding angles that are congruent, then the lines are parallel.